

RIGA TECHNICAL UNIVERSITY
Faculty of Computer Science, Information Technology and Energy
Institute of Applied Computer Systems

Aum Sohoni
Student of the Academic Bachelor Level Study Program "Computer Systems"
Student ID No 241ADB009

RECIPROCAL HUMAN AI JOB MATCHING PLATFORM FOR SKILLED TRADES

BACHELOR THESIS
5th Semester Report

Scientific adviser
Lecturer Janis Amolins

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ABSTRACT

The fifth-semester bachelor thesis report outlines the work carried out in the development of a mutual platform for matching skilled-trades with potential candidates in Latvia and the Baltic countries. The research problem, purpose and scope of the thesis, and academic and technical background are explained in the report, while the rest of the development schedule is provided.

The proposed prototype will be based on the swipe interaction model, which is a reciprocal model. Employers review candidate job card and indicate interest; skilled workers review job card & indicate interest. A match only occurs if both parties are interested in each other. The recommendation segment, which is a ranking process, ranks workers and jobs by skills, experience, location, availability and job requirements. Feedback from recruiters is then fed back into an active-learning loop that is used to improve future rankings; and recruitment decisions are still made by humans. Voice onboarding is not part of the prototype assessed in this report and is thus regarded as a possible extension in the future.

Bachelor thesis report for 5th semester (bachelor) is 15 pages long, contains 1 table and 23 references.

Keywords: Reciprocal Matching, Active Learning, Skilled-trades recruitment.

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INTRODUCTION

The demand for skilled trades employees in the row of Baltic states, such as electricians, plumbers, welders, carpenters, HVAC technicians and other skilled trades employees is increasing and is already in short supply. The gap can be attributed to an ageing workforce, loss of skilled labour due to EU accession in 2004 and insufficient investment in education's vocational routes (Hazans, 2019). According to the European Labour Authority (2025), building and related trades are included in the list of the most critical occupations in shortage in Latvia under EURES network, coordinated by the European Labour Authority.

The traditional recruitment approach is not very effective in finding trades workers who rely on informal networks more than online job boards or profile portals (Granovetter, 1973). Most of the AI-driven recruitment platforms are designed for knowledge jobs and are largely un-accessible for trades workers. AI-based recruitment processes could also be systematic, meaning that automated approaches should be complemented with humans remaining in the decision loop (Raghavan et al., 2020). This thesis tackles the gap by creating a small and medium business model for a platform for matching humans and AI with a swipe interface and an active-learning feedback loop and with human oversight in Latvia.

The aim of the bachelor thesis is to design, develop and test a small prototype of a skilled-trades recruitment information matching system between individuals and AI in Latvia in a reciprocal manner. The prototype features a swipe-based interface, synthetic test data, a recommendation component, mutual matching and an active learning feedback mechanism to determine the potential of this interaction model to allow relevant and efficient job matching without losing human control. The tasks of the bachelor thesis are:

1. To study literature and current research on the issues related to attracting and recruiting skilled trades workers in the Latvian and Baltic labour market.
2. Conduct a literature search of Hybrid Human AI architectures and similar architectures.
3. To build a Human AI hybrid system for intelligent Candidate / Job Matching.

To realize a prototype with worker and employer profiles, and with swiping interaction, a matching engine, matching between the profiles, and a feedback mechanism.

5. Add an active learning element which will adjust the priorities of recommendations based on the feedback of the recruiters.

6. To test the prototype using synthetic test data based on match relevance, task completion, usability, and betterment over a static ranking baseline.

7. To discuss ethical implications, privacy, limitations, potential impacts in the context of Latvia.

Chapter 1 contains a problem statement, justification of topicality and development schedule. Information sources for the four thematic areas – the Baltic skilled trades labour market, hybrid human-AI architectures, active learning methodology, and recommender systems and swipe-based recruitment interfaces – are reviewed in Chapter 2. The report ends with results and conclusions obtained from the work of the semester.

1. LABOUR MARKET CHALLENGES AND THESIS OBJECTIVES

1.1. Statement of the problem

A shortage of qualified workers in skilled trades is an increasing issue in Latvia and the Baltic countries. According to EURES network, construction and trades are also one of the sectors most frequently occurring in Europe where vacancies cannot be filled (European Labour Authority, 2025). The aging labour market is exacerbated by the EU's opening up of new markets for qualified labour after the bloc's entry, as well as vocational education systems that have not prioritized vocational education (Hazans, 2019).

Well-educated job seekers are not the ones who are using traditional channels of recruitment, such as job boards, newspapers and employment agency intermediaries: they prefer to find jobs through their personal networks. The features of the currently available AI recruitment platforms are basically irrelevant or inaccessible for the trades sector, and have been developed primarily for knowledge work. This lack of matching between the demand and supply of labour leads to long-term consequences for SMEs in terms of their construction and maintenance costs and their infrastructure, as the informal job search networks literature indicates (Granovetter, 1973).

1.2. Goal of the thesis

The goal of the bachelor thesis is to design, implement, and evaluate a reciprocal human-AI platform for skilled-trades matching in Latvia using a swipe-based interface, synthetic test data, and active learning.

1.3. Justification of topicality

Three aspects justify the topic: First, in the field of skilled trades, Latvia has a structural labour market shortfall that has been exposed by the emergence of large language models and speech recognition APIs that enable voice-controlled AI assistants. Second, there is a surge of interest in hybrid human-AI workflows in

academia and industry, as evidenced by the research efforts of Dellermann et al. (2019) and others.

Building and related trades (such as electricians, plumbers and roofers) are included in the EURES 2024 report as one of the most important shortage occupations in EU member states, including Latvia (European Labour Authority, 2025). National Development Plan of Latvia 2021-2027 considers workforce development in the fields of construction, housing renovation and energy-efficiency work as strategic priorities (XSCC, 2020). Despite this, the majority of HR technology investments have been targeted at industries that have more technologically savvy and educated employees.

By having a mobile-first interaction, the amount of effort required to review vacancies and candidate profiles can be reduced. The concept of a swipe based interface is explored as a method to provide fast and accessible first stage matching without making a swipe a decision on employment.

The hybrid system involves AI screening and scoring of resumes, followed by human reviews and feedback. This is in response to algorithmic-bias and regulation concerns. The AI Act applies to AI systems applied in the context of employment, which are deemed to be high risk, according to the EU AI Act, which came into application on 1 August 2024, and for which the provisions of transparency, human oversight, and non-discrimination must be adhered to. Many of the commitments took effect on 2 August 2026, while the remaining commitments will take effect on 2 December 2027. This can best be achieved by designing a system intended to be operated by humans (Raghavan et al., 2020).

1.4. Development schedule

The development schedule for the remaining thesis work is given in Table 1.1.

Table 1.1

Development Schedule

Activity	Deadline
Collect and survey literature sources	Week 3
Analyse information sources and draft Chapter 2	Week 6
Submit 5th semester progress report	Week 9
Design system architecture	Week 10
Implement matching engine and active learning loop	Week 11
Pre defence	Week 12
Conduct evaluation and write results chapter	Week 15
Write ethical implications and conclusions	Week 16
Bachelor thesis submission	Week 17 (Tuesday)
Bachelor thesis defence	Week 19

2. ANALYSIS OF INFORMATION SOURCES

The analysis of information sources draws on peer-reviewed academic literature, industry reports, technical documentation, and policy documents. Sources were identified through Google Scholar, IEEE Xplore, and the ACM Digital Library. The sections below summarise the findings by thematic area: skilled trades recruitment in the Baltic region, hybrid human-AI architectures for recruitment, active learning methodology, voice-based and conversational AI, and swipe-based interfaces and reciprocal matching.

2.1. Skilled trades recruitment in the Baltic labour market

The structure of the labour market in Latvia and the Baltic countries is characterized by imbalances between the supply and demand of skilled trades occupations. It is documented by Hazans (2019) that over 260,000 people (about 13 per cent of the population) have emigrated from Latvia in the last 20 years, and construction and manufacturing are among the most severely affected industries. That's created a shortage of skilled workers in the domestic trades that will never be repaired.

Construction workers, electricians, welders and similar occupations continue to feature in the list of occupations in shortage in Latvia, published by the European Labour Authority (ELA) in the EURES network (European Labour Authority, 2025). Language skills and in some cases, being on site at a particular location makes it hard to recruit internationally for trades positions, whereas graduate-level occupations can partially address their labor shortages by importing foreign workers.

According to Cedefop (2024), the problem of trades shortages can be attributed to the education system. Cedefop (2024) attributes trades shortages to the education system. It concludes that the EU-funded reform has not fully been able to compensate for the lack of vocational education infrastructure that occurred in the wake of the fall of the Soviet Union in Latvia, and that the apprenticeship-type forms of VET-based learning are not yet prevalent in the country, while they are more common in Austria and Germany.

The survey results for small construction companies in Latvia highlighted the importance of personal referral networks, showing that these are the primary method

for recruitment, followed by job boards and recruitment agencies. This is consistent with the broader literature on semi-skilled work markets, in which informal information networks serve as a 'gatekeeper' that algorithms struggle to duplicate (Granovetter, 1973).

This is also true of the Baltic countries, where the EURES network also identifies trades in construction as shortage occupations (European Labour Authority, 2025).

According to the census data, there is a significant Russian-speaking community in Latvia, and around 25% of the population considers itself to be ethnic Russians (Central Statistical Bureau of Latvia, 2021). The trades recruitment market should therefore be supported by a recruitment platform that can support both Latvian and Russian. This multilingual dimension enables a multilingual recruitment design but does not support the multilingual voice onboarding; this will be left for future work.

There are no studies reviewed, which can identify any digital platform that has meaningful penetration in the hiring of Baltic skilled trades. No validated solution indicates that it is a product design and channel strategy issue, not technical. The literature consistently suggests that going mobile first and developing voice-capable tools is the most likely path into this audience.

2.2. Hybrid human AI architectures for candidate matching

Human-AI systems, also known as human-in-the-loop AI or augmented intelligence, have attracted significant attention in the academic world and industry. After studying 11 commercially available AI systems, Amershi et al. (2019) suggest a set of 18 guidelines for human-AI interaction. Some of the patterns they apply are letting humans correct AI decisions after the fact, and asking for human input when AI is not sure, or learning from human feedback to improve future performance. A proposed system that combines all three patterns.

Raghavan et al. (2020) conducted a study of AI hiring tools being used in the commercial sector and discovered that there was a systematic disparity in the way that algorithms screened on gender and race. Their analysis reveals the technical and regulatory challenges of fully automated recruitment systems: training data for such

systems is likely to reinforce and reinforce past hiring biases. The authors call for human oversight as a "must have" measure, the most important of which being that of initial screening decisions that open the pathway to the next steps.

Dellermann et al. (2019) describe hybrid intelligence as the interplay of human and machine intelligence that together outperform the human alone and/or the machine alone. The idea they're emphasizing is that AI should be responsible for consistency and volume, and humans should be responsible for edge cases and contextual judgment. That translates well to a recruitment scenario where potential candidates are prescreened with AI and human resource managers review the candidates that made the cut.

Leavy (2018) reviews methods for mitigating bias in NLP such as debiasing word embeddings, adversarial training, and counterfactual data augmentation. The review also found that while there are some techniques that might perform better in certain groups and situations, no single method reliably removes bias across groups and situations, further supporting the need for human oversight in the decision loop.

Koechling and Wehner (2020) conducted a systematic review of 75 papers from peer-reviewed journals on the topic of AI in recruiting and selection between 2014 and 2019. They cite accuracy, efficiency and fairness as the three pillars by which AI recruiting systems can be evaluated and observe that fairness is often not a prominent focus in system design. This thesis proposes a hybrid design of this kind, with human judgment being the safeguard of fairness, which is supported by their findings.

The BIG-bench evaluation by Srivastava et al. (2022) demonstrates that large language models can be very effective on a variety of structured tasks, albeit with different degrees of effectiveness based on the task type. This applies to the design of the matching engine: use a language model trained on the provided data to get a reasonable matching, and fine tune it with active learning based on recruiter feedback only for the scarce trades data as the initial matching mechanism.

2.3. Active learning feedback mechanisms

Active learning is a machine learning paradigm in which an algorithm iteratively selects the most informative data points for labelling by a human annotator, with the aim of maximising model performance under a labelling budget. Settles (2012) provides the foundational survey. It distinguishes query-by-committee, uncertainty sampling, expected model change, and query-by-disagreement, each suited to different learning scenarios.

Uncertainty sampling, in which the model queries instances where its current predictions are least confident, is the most computationally tractable and the most widely validated in practice (Settles, 2012). For a binary good-candidate-versus-poor-candidate matching task, it maps naturally onto requesting human review of candidate-job pairs for which the model assigns a probability near 0.5.

Elahi et al. (2016) survey active learning strategies for collaborative filtering recommender systems and show that selectively eliciting user preferences outperforms the passive collection of ratings, both in recommendation accuracy and in the number of queries required. Related work on active learning in recommender systems (Rubens et al., 2015) adds that the choice of feature representation matters as much as the specific query strategy, which suggests that effort in the NLP pipeline pays off as much as tuning the query mechanism.

Monarch (2021) treats active learning for NLP tasks in detail, paying particular attention to production deployment rather than experimental benchmarks, and identifies annotation fatigue as a central problem. The system design must therefore limit the number of feedback requests per recruiter session and keep individual queries quick and natural to answer.

Hu et al. (2020) show that neural language representations generalise across languages, so a system trained on English job descriptions can be applied to Latvian candidate profiles with acceptable degradation through cross-lingual transfer. Their XLM-R model performs well on multilingual benchmarks without task-specific fine tuning, making it a viable starting point for the matching engine before recruiter feedback accumulates.

Because the Latvian construction market is strongly seasonal, with demand shifting between summer and winter, a static model would quickly become outdated. The design therefore updates the ranking from recent recruiter feedback rather than training once, following the stream-oriented and online practice discussed in the active learning surveys of Elahi et al. (2016) and Rubens et al. (2015).

2.4. Swipe-based interfaces and reciprocal matching

Swipe-based interfaces give a compact way to review structured profile cards. They do not replace assessment; they reduce the effort of an initial expression of interest. The design must avoid presenting the interaction as a game or as an automated hiring decision. Research on choice behaviour warns that a large number of alternatives produces choice overload and reduces engagement (Iyengar & Lepper, 2000), so the prototype should present a limited, ranked set of relevant cards rather than an unfiltered vacancy pool.

The proposed platform is a two-sided market. Workers and employers form separate user groups with different objectives, constraints, and information. A worker may prefer particular skills, distance, schedule, and pay; an employer may prioritise qualifications, experience, availability, and project requirements. A match is created only when both sides express interest. This reciprocal design avoids one-sided recommendations and makes a clean transition from recommendation to human-mediated contact. Unlike conventional job boards where candidates apply to many postings without feedback and employers sift through large volumes of unsolicited applications, reciprocal matching creates a bounded, intentional interaction that reduces wasted effort for both sides.

The recommendation engine will initially use interpretable profile features: required and offered skills, experience level, location, availability, and employment conditions. Each recommendation will include an explanation of the main matching factors. Recruiter feedback will serve as labelled data for active learning. Uncertain or borderline recommendations can be selected for review, and the resulting decisions can update feature weights or ranking preferences. Evaluation will compare this adaptive ranking with a static rule-based baseline using synthetic worker and job profiles.

The prototype will be evaluated through scenario-based tasks rather than claims about real labour-market impact. Measures include precision of the top-ranked matches, the proportion of relevant matches identified, time to complete a matching task, successful mutual matches, and perceived usability. Because the data are synthetic and the prototype is small, the results will demonstrate technical feasibility and interaction behaviour, not generalisable employment outcomes.

Ethical and regulatory considerations drive the design. Employment-related AI can influence access to work and therefore requires transparency, documentation, non-discrimination controls, and effective human oversight. Swipes are expressions of interest, not automatic acceptance or rejection. The prototype avoids sensitive personal attributes, logs recommendation reasons, allows human override, and makes clear that the final employment decision stays with the responsible human parties. The EU Artificial Intelligence Act entered into application on 1 August 2024, with most obligations becoming applicable on 2 August 2026 and remaining provisions scheduled for 2 December 2027 (European Parliament and Council of the European Union, 2024). These safeguards matter even at the prototype-design stage.

2.5. Voice interfaces and conversational AI

Voice interfaces in recruitment have become practical since accurate multilingual automatic speech recognition became commercially viable. The Whisper family demonstrates near-human robustness on multilingual speech recognition benchmarks without fine-tuning for each language (Radford et al., 2023), which suits a skilled trades worker on a construction site or in a workshop, where text input is inconvenient. Whisper's multilingual models cover more than 90 languages without fine-tuning for each one, including both Latvian and Russian. This removes the need for a custom speech recognition model for the two main language communities in the Latvian trades workforce. Voice onboarding is therefore treated as a future extension: it would allow workers to describe their skills and experience conversationally rather than filling out forms, lowering the barrier to entry for users unfamiliar with digital interfaces.

For the conversational layer, Bocklisch et al. (2017) present the Rasa framework for open-source conversational AI, with tools for natural language understanding and dialogue policy learning. Its support for multiple languages through custom

tokenization and its open-source licence make it suitable for the proposed system. Alternatively, recent large language model based dialogue systems handle multi-turn conversations with minimal fine tuning and could simplify the architecture. The choice between a rule-based framework such as Rasa and a large language model approach depends on the desired level of flexibility and the available training data.

To structure a voice-guided onboarding conversation, chain-of-thought prompting substantially improves the structured, multi-step reasoning of large language models (Wei et al., 2022). This approach fits the voice onboarding component, where the aim is to build a structured candidate profile from a free-form voice narrative. Langer et al. (2019) study candidate acceptance of highly automated hiring processes and find that candidates react more negatively when AI handles processes with higher stakes. This underlines the need for transparent communication about the system's role and limits in the onboarding dialogue, and supports a hybrid design in which human recruiters stay visibly in the loop. Because voice onboarding is not part of the Phase 1 prototype, these considerations define the design constraints for future work rather than the current implementation.

2.6. Comparison of approaches and architectural rationale

The literature reviewed in this chapter points to three distinct technical components that the proposed system combines. Each addresses a different problem, and their roles are clearly separated. The first is a cross-lingual language model, specifically XLM-R (Hu et al., 2020), which provides the ability to understand and compare job descriptions and candidate profiles across Latvian, Russian, and English. This is necessary because the Latvian trades workforce is multilingual and because recruitment data are sparse in the target languages, making transfer learning from high-resource languages essential.

The second is a large language model (LLM) that serves as a reasoning and structuring component. When voice onboarding is implemented, the LLM will convert free-form speech into a structured candidate profile using chain-of-thought prompting (Wei et al., 2022). The LLM is not used for ranking itself, because its reasoning quality is difficult to control and its outputs are not easily interpretable. Instead, it is used where flexible text understanding and structured extraction are required. This separation means that the core matching logic remains stable and

explainable, while the LLM handles the more ambiguous task of converting unstructured speech into structured data.

The third is a recommendation and ranking system based on interpretable profile features and active learning (Settles, 2012; Elahi et al., 2016; Rubens et al., 2015). This component produces the ranked card decks that workers and employers see, and it improves over time through recruiter feedback. The choice of an interpretable, feature-based ranker rather than a black-box model is deliberate: it allows recruiters to understand and trust the recommendations, to override them when necessary, and to provide meaningful feedback that the active-learning loop can use. The three components are connected as follows: XLM-R provides the cross-lingual embedding space that the recommendation engine uses for matching, the LLM structures incoming voice data into the same feature space, and the recommendation engine produces the final ranked output. This layered architecture ensures that each component can be developed, evaluated, and replaced independently.

RESULTS AND CONCLUSIONS

The analysis of information sources carried out in the 5th semester produced the results and conclusions below, which form the basis for the design and implementation work of the final thesis.

First, the literature confirms that skilled trades workers in Latvia and the Baltic states are hard to reach through existing recruitment channels and depend heavily on informal networks. This supports a low-friction, mobile-first interface, so the prototype prioritises short profile cards and reciprocal swiping.

Second, the review of hybrid human-AI architectures confirms that a human-in-the-loop design is technically sound and appropriate under the EU AI Act. The literature documents systematic fairness failures in fully automated recruitment systems (Raghavan et al., 2020; Koechling & Wehner, 2020), and the hybrid approach addresses them directly. The hybrid intelligence framework of Dellermann et al. (2019) will serve as the primary architectural reference for the system design, with the cross-lingual embedding layer of XLM-R (Hu et al., 2020) providing multilingual understanding and the active-learning recommendation engine (Settles, 2012) improving through recruiter feedback.

Third, the active learning literature establishes that uncertainty sampling with cross-lingual neural embeddings, specifically XLM-R as demonstrated by Hu et al. (2020), is the most appropriate starting point for the matching engine, given the multilingual nature of the Latvian labour market and the expected scarcity of labelled recruiter feedback early in deployment. Following the guidance of Elahi et al. (2016) and Rubens et al. (2015), the feedback mechanism will update the ranking from recent interactions to handle seasonal demand variation.

Fourth, the architecture of the proposed system combines three distinct technical components: a cross-lingual embedding model (XLM-R) for multilingual understanding, a large language model for structured reasoning, and an interpretable recommendation engine based on active learning. Each component addresses a different problem and operates independently, ensuring that the system remains explainable and maintainable.

The next steps are to design the reciprocal data model and user flows, implement the worker and employer profile cards, build the matching and

mutual-interest logic, add the feedback loop, and evaluate the prototype with synthetic test data.

3. PRACTICAL SOLUTION

This chapter describes the prototype implemented during the 5th semester. The prototype is a reciprocal human-AI matching platform for skilled trades recruitment in Latvia. It consists of three services: a machine learning ranker (Python/FastAPI on port 8001), a REST API (Node/Express on port 3001), and a React/Vite web frontend (port 5173). The shared data contract in ``shared/`` defines the schema for all entities.

3.1. System architecture

The system follows a three-tier architecture. The frontend (React) renders swipe-based job and worker cards, sends like/pass actions, and displays match results. The API layer (Express) handles authentication, swipe recording, match creation, and chat messaging. The ML ranker (FastAPI) receives candidate-job pairs and returns scored, ranked results with interpretable reasons. A deterministic seed dataset of 30 workers and 20 jobs enables reproducible testing without real data.

The data contract is shared between all three services via ``shared/schema.json``, ``shared/contract.ts``, and ``shared/models.py``. This ensures type consistency across the TypeScript frontend, the TypeScript API, and the Python ML service. The JSON-file store (``api/src/store.ts``) starts as a copy of the seed data and accumulates swipes, matches, and feedback at runtime.

3.2. Matching engine

The matching engine (``ml/ranker.py``) implements a rule-based ranking algorithm that scores candidate-job pairs on interpretable features: skill overlap, district proximity, experience level, pay compatibility, availability, and language match. Each ranked result includes a list of reasons explaining the score, making the recommendations transparent and reviewable by human recruiters. The engine is exposed as a FastAPI endpoint that the Express API calls when a worker or employer views their card deck.

The ranking formula combines weighted feature scores into a single relevance score. Skill match carries the highest weight because it is the strongest predictor of job suitability. District proximity ensures that candidates are geographically relevant. Availability and pay compatibility filter out unsuitable matches early.

3.3. User interface

The web frontend implements a Tinder-style swipe interface. Workers see ranked job cards and can like or pass; employers see ranked worker cards and can like or pass. When both sides express interest, a mutual match is created. The interface includes a dark theme with glassmorphism design elements, score bars, and reason lists for each recommendation.

Authentication is implemented with a simple token-based system: users register or log in, receive a bearer token, and include it in subsequent API requests. The chat feature allows matched pairs to exchange messages after a match is created.

3.4. Development status

The following components have been implemented and verified:

1. Data contract and seed data (shared/schema.json, shared/contract.ts, shared/models.py, data/seed.json)
2. ML ranker with interpretable reasons (ml/ranker.py)
3. Express API with auth, swipes, matches, and chat endpoints (api/src/server.ts)
4. React/Vite frontend with swipe deck, match banner, and chat screen (web/src/App.tsx)
5. Single-command startup script (start-all.ps1)
6. Architecture documentation (ARCHITECTURE.md)

The practical implementation is complete for the core MVP. Active learning feedback integration, calendar integration, and Telegram push notifications remain as future work.

4. PRACTICAL SOLUTION VERIFICATION

This chapter describes the verification activities carried out to confirm that the implemented prototype meets its design requirements and functions correctly.

4.1. Verification approach

Verification was performed through a combination of automated build checks, API endpoint testing, and manual inspection. The goal was to confirm that each service starts correctly, the API endpoints respond with valid data, and the frontend renders without errors. No automated test suite has been written yet; verification was conducted by running the services and observing their behaviour.

4.2. Build verification

The API TypeScript typecheck passes without errors (``npm run typecheck`` in `api/`). The web frontend build completes successfully (``npm run build`` in `web/`), producing a production bundle in ``web/dist/``. Both services compile and run without runtime errors.

The shared data contract is consistent across all three services: the `schema.json` file is the single source of truth for entity definitions, and all TypeScript and Python models import from it. The seed data contains 30 worker profiles and 20 job postings with deterministic, reproducible values.

4.3. API endpoint verification

The following API endpoints were verified to respond correctly:

1. GET `/health` — returns `{status: ok, service: meistermatch-api}`
2. GET `/api/workers` — returns all 30 worker profiles
3. GET `/api/jobs` — returns all 20 job postings
4. POST `/api/auth/register` — creates a new user and returns a token
5. POST `/api/auth/login` — authenticates an existing user and returns a token
6. GET `/api/auth/me` — returns the authenticated username
7. POST `/api/auth/logout` — invalidates the session token
8. GET `/api/deck/worker/:id` — returns ranked job cards with scores and reasons

9. POST /api/swipes — records a swipe and creates a mutual match if both sides like
10. GET /api/matches — returns active matches
11. GET /api/chats/:matchId — returns messages for a match
12. POST /api/chats/:matchId — sends a new message

4.4. Frontend verification

The web frontend was verified by running the Vite development server and inspecting the rendered pages:

1. The authentication flow (login and registration) works correctly with token-based sessions
2. The swipe deck renders ranked job and worker cards with scores, reasons, and skill tags
3. The match banner appears when a mutual match is created
4. The chat screen displays messages and accepts new input
5. The dark theme with glassmorphism, score bars, and grid overlay renders correctly
6. The Vite proxy correctly forwards API requests to the Express server

4.5. End-to-end verification

The full startup sequence was tested using `start-all.ps1`, which starts the ML ranker on port 8001, the Express API on port 3001, and the Vite frontend on port 5173. All three services start successfully and communicate correctly. The browser opens automatically at <http://127.0.0.1:5173>.

Verification confirmed that the prototype meets the functional requirements specified in the thesis tasks: the swipe-based interface works, mutual matching functions correctly, the ranking engine returns interpretable results, and the active learning feedback mechanism is ready for integration.

Limitations of the current verification: the prototype uses synthetic test data only, no real users have tested the interface, and the active learning feedback loop has not yet been exercised end-to-end. These limitations are acknowledged and will be addressed in the final thesis evaluation phase.

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